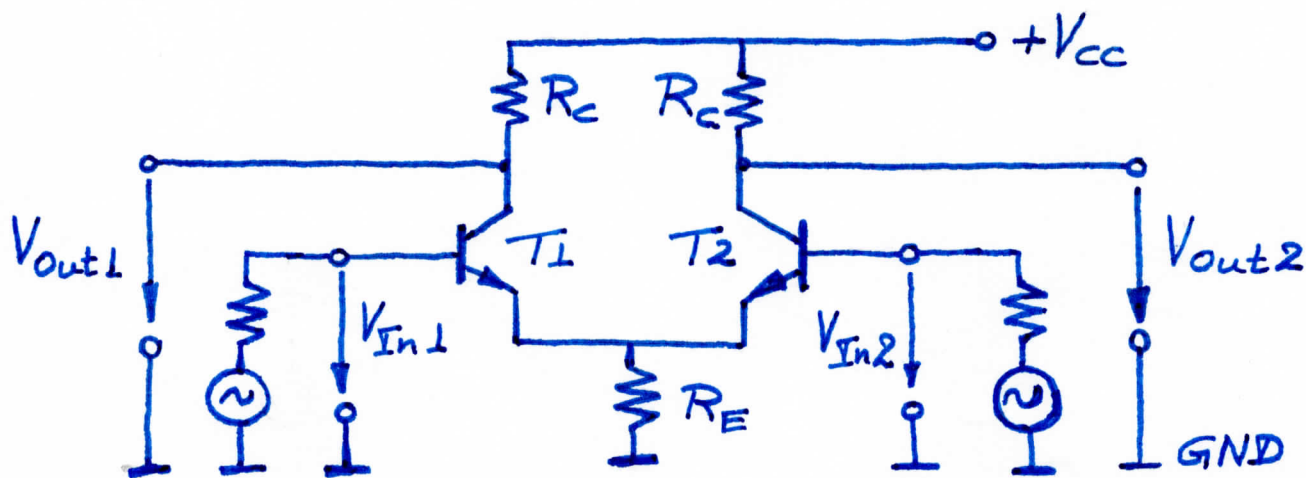


Iconic circuit: Differential amplifier

- * Differential amplifier is physical basis of Operational Amplifier (Op Amp).
- * Two inputs. Two outputs.
- * Amplifies difference in input voltages.

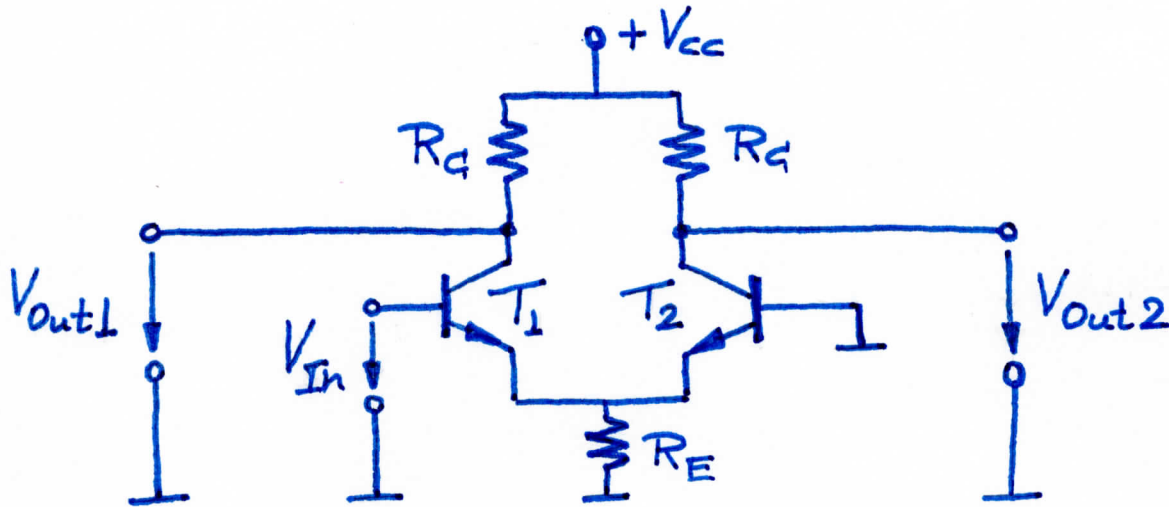


- * Emitters are coupled and share R_E .
- * 1 or 2 inputs (unused input can be grounded)
- * 1 or 2 outputs (unused output can be floating)
- * Two modes of operation

Common mode: $V_{In1} = V_{In2}$

Differential mode: $V_{In1} \neq V_{In2}$
→ Difference $\neq 0$

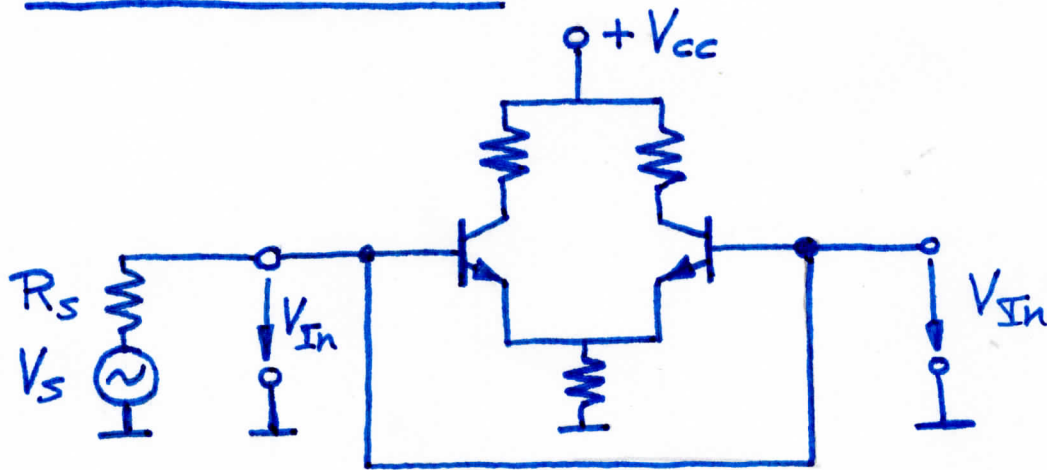
Inverting output and non-inverting output ^②



Q: Which output is inverting output?
 $\Rightarrow$ Output 1 is inverting.

Q: Which output is non-inverting output?
 $\Rightarrow$ Output 2 is non-inverting.

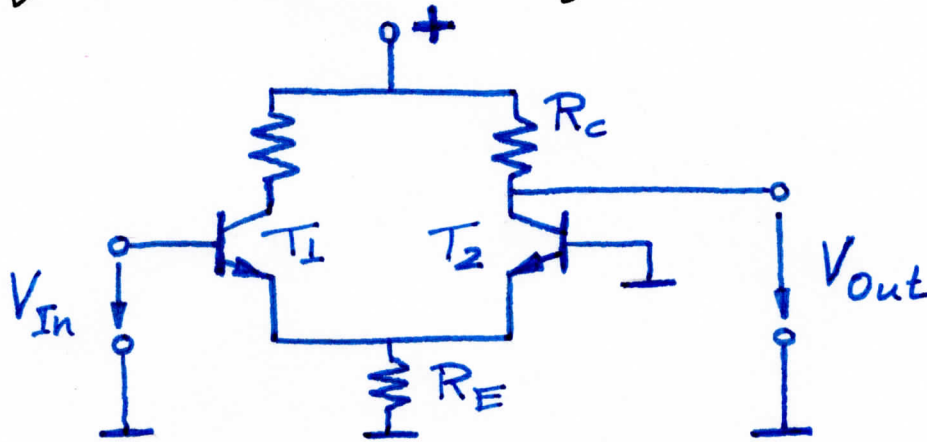
Common mode (CM): Both inputs at same voltage



$$\Rightarrow V_{In1} = V_{In2} = V_{In}$$

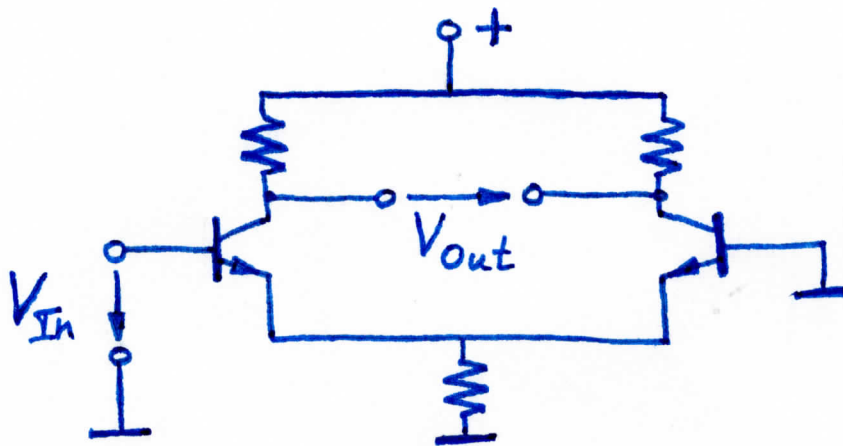
Input and output configurations

* Single input & single output



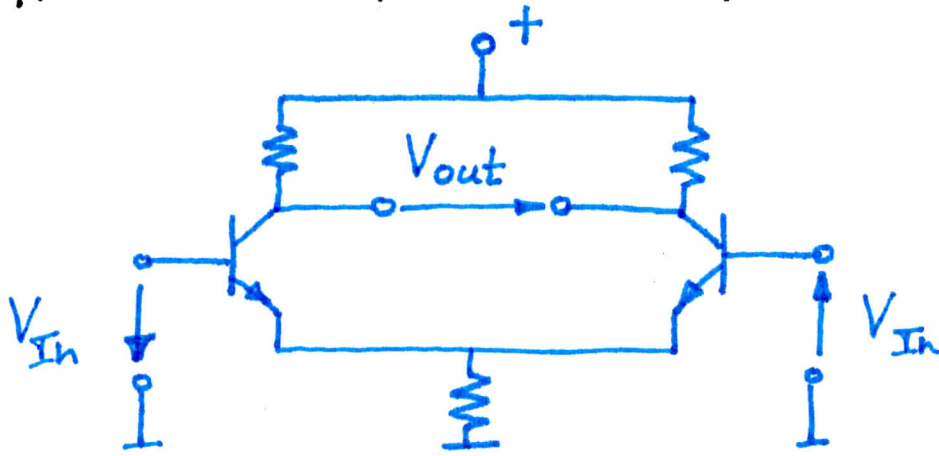
⇒ To be shown: $A = R_C / 2r_E$

* Single input & differential output



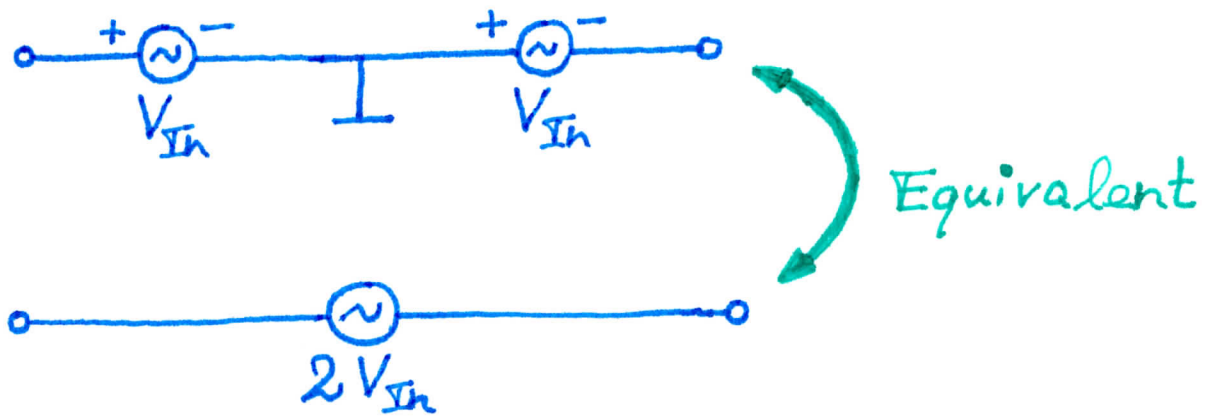
⇒ To be shown: $A = R_C / r_E$

* Differential input & differential output

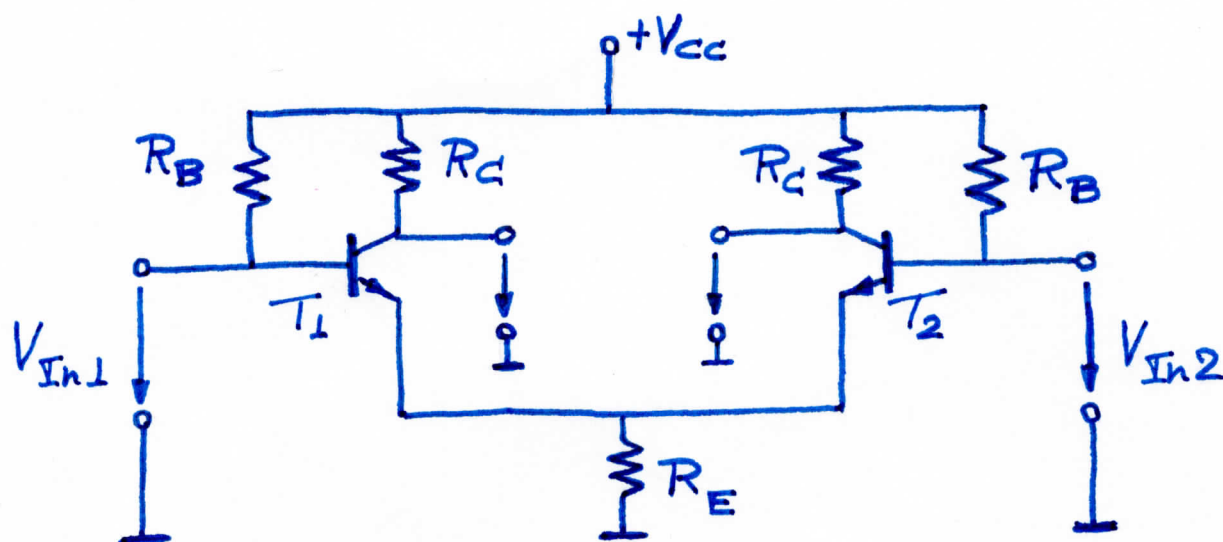


⇒ To be shown: $A = \frac{2R_c}{r_E}$

Note on input voltage representation



Quiescent point (Q-point)



⇒ Symmetric circuit

⇒ Q-point: $I_{B1} R_B + V_{BE} + 2(\beta+1) I_{B1} R_E = V_{CC}$

$$\Rightarrow I_{B1} = \frac{V_{CC} - 0.7V}{R_B + 2(\beta+1) R_E}$$

$$\Rightarrow I_{E1} = (\beta+1) I_{B1} \quad I_{C1} = \beta I_{B1}$$

$$\Rightarrow V_E = 2 I_{B1} (\beta+1) R_E$$

$$V_C = V_{CC} - \beta I_{B1} R_C$$

$$\begin{aligned} \Rightarrow V_{CE} &= V_C - V_E = V_{CC} - \beta I_{B1} R_C - 2 I_{B1} (\beta+1) R_E \\ &\approx V_{CC} - \beta I_{B1} (R_C + 2 R_E) \end{aligned}$$

... this concludes our Q-point calculation.

Q: What would be a good value of V_{CE} ?

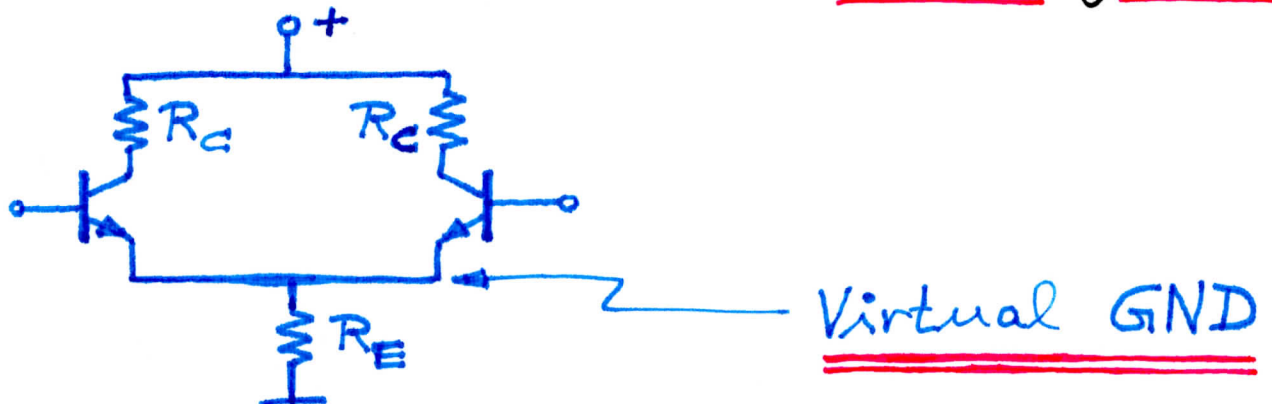
Qualitative discussion

⑥

* Common mode : $V_{In1} = V_{In2} \Rightarrow$ Due to symmetry of circuit $I_{C1} = I_{C2} \Rightarrow V_{out} = 0 \Rightarrow \underline{\underline{A_{CM} = 0}}$

* Differential mode : Assume $V_{In1} \uparrow \Rightarrow \Rightarrow V_{BE1} \uparrow$ and $V_E \uparrow \Rightarrow V_{BE2} \downarrow \Rightarrow$
 $\xrightarrow{\sim 0.7V} \quad \xrightarrow{\sim 0.7V}$
 $\Rightarrow I_{E1} \uparrow$ and $I_{E2} \downarrow \Rightarrow I_{C1} \uparrow$ and $I_{C2} \downarrow$
 $\Delta I_{C1} = \Delta I_{C2}$ due to linearization of circuit.
 $\Rightarrow$ Current is "steered" through T_1 & T_2

* $I_{RE} = I_{E1} + I_{E2}$. Since $I_{E1} \uparrow$ and $I_{E2} \downarrow$ (by same amount) $\Rightarrow I_{RE} = \text{constant} \Rightarrow V_{RE} = \text{constant} \Rightarrow V_E = \text{constant} \Rightarrow$ Emitters are at constant potential $\Rightarrow \underline{\underline{"virtual ground"}}$



Amplification

$$A_{CM} = \frac{V_{out}}{\frac{1}{2}(V_{in1} + V_{in2})} = 0$$

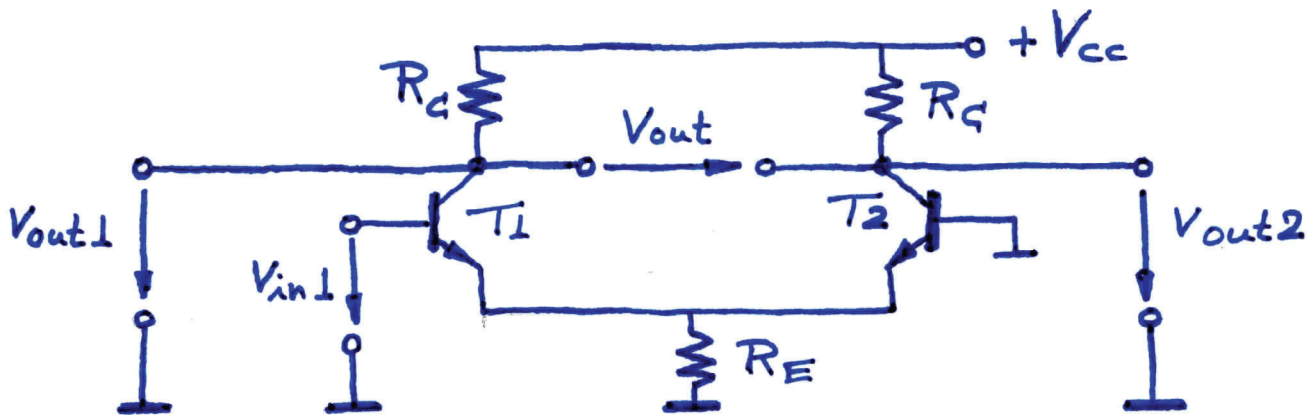
(already shown)

Single input & single output $A = \frac{V_{out1}}{V_{in1}} = ?$

Single input & differential output $A = \frac{V_{out1} - V_{out2}}{V_{in1}} = ?$

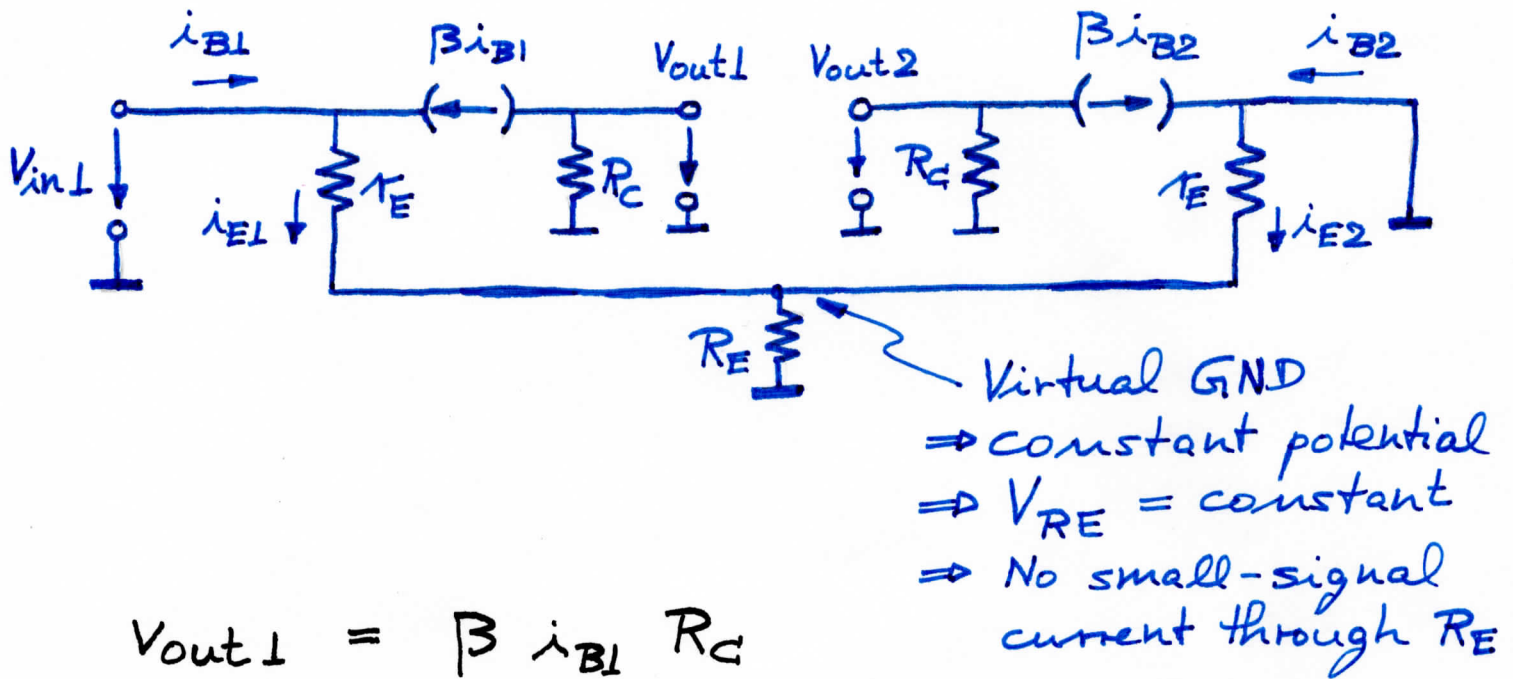
Differential input & output $A = \frac{V_{out1} - V_{out2}}{V_{in}} = ?$

Next, we consider single input & differential output
 $\Rightarrow$ Circuit diagram



Small-signal equivalent circuit

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$$V_{out1} = \beta i_{B1} R_C$$

$$V_{out2} = \beta i_{B2} R_C$$

$$V_{in1} = i_{E1} r_E + i_{E1} r_E = 2 i_{E1} r_E$$

$\xrightarrow{\text{blue}} = -i_{E2}$

$$A = \frac{V_{out1} - V_{out2}}{V_{in1}} = \frac{\beta i_{B1} R_C - \beta i_{B2} R_C}{2 i_{E1} r_E}$$

$\xrightarrow{\text{blue}} = -i_{B1}$

$$= \frac{2 \beta i_{B1} R_C}{2 (\beta + 1) i_{B1} r_E} \approx \underline{\underline{\frac{R_C}{r_E}}}$$

* Single input & single output: $A = \underline{\underline{\frac{R_C}{2 r_E}}}$ Why?

* Differential input & output: $A = \underline{\underline{\frac{2 R_C}{r_E}}}$ Why?

Summary table

Input	Output	Gain
single	single	$R_C/2r_E$
single	differential	R_C/r_E
differential	differential	$2R_C/r_E$

Example: Consider a differential amplifier with differential input & output. $V_{CC} = 12V$. Assume $R_C = 500\Omega$ and $I_{C1} = 10mA$.

* Determine the voltage gain. $I_{E1} \approx I_{C1} = 10mA$

$$\Rightarrow r_E = \frac{V_t}{I_{E1}} = \frac{26mV}{10mA} = 2.6\Omega$$

$$\Rightarrow A = \frac{2R_C}{r_E} = \frac{1000\Omega}{2.6\Omega} = \underline{\underline{384}}$$

* What is the gain of a 2-stage amplifier?

$$\Rightarrow A = A_{1st-stage} \times A_{2nd-stage} = 384 \times 384 = \underline{\underline{148000}}$$

$\Rightarrow$ Reasonable value for Op Amp.

Applications

- * Operational amplifier. The basic internal circuit of an Op Amp is a 2-transistor differential amplifier. Amplification of Op Amp $\approx 100,000$. Amplification of differential amplifier ≈ 200 . $\Rightarrow$ Multiple stages needed.
- * Emitter-coupled logic (ECL). Also called current steering logic (CSL).
ECL is very fast. But: Power consumption is high. ECL consumes power at the quiescent point (Q-point).